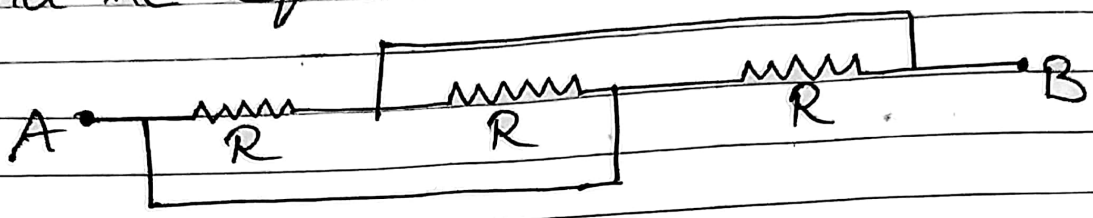
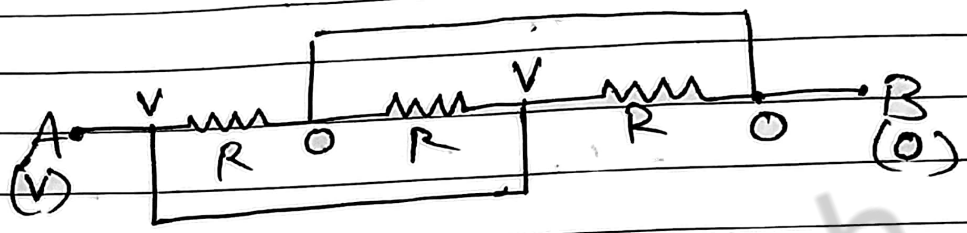
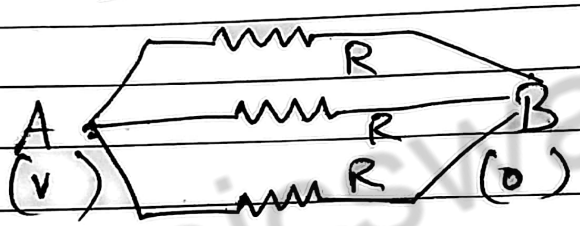
Combination of Resistors – Part II – Complex

I) Potential Method

Find the equivalent Resistance between A & B.

① 

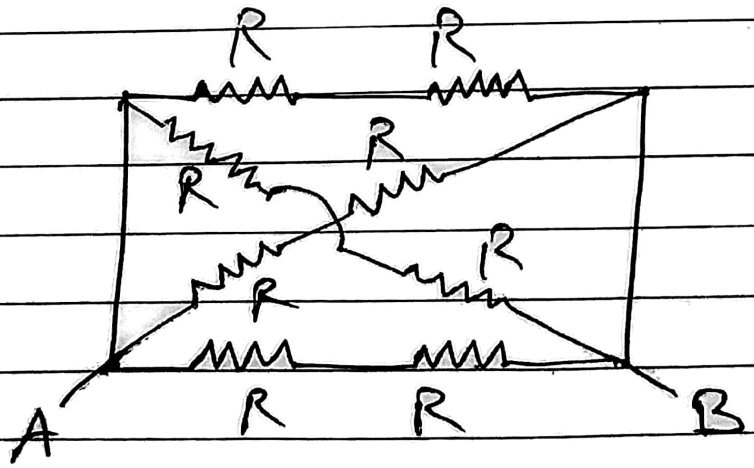
Solution 



$$\frac{1}{Re} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

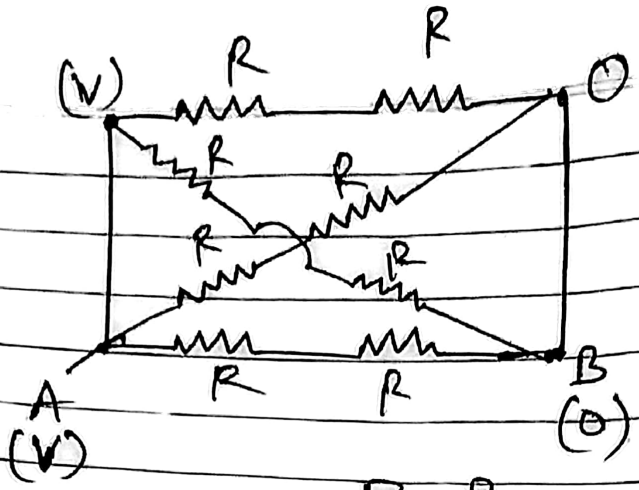
$$Re = \frac{R}{3}$$

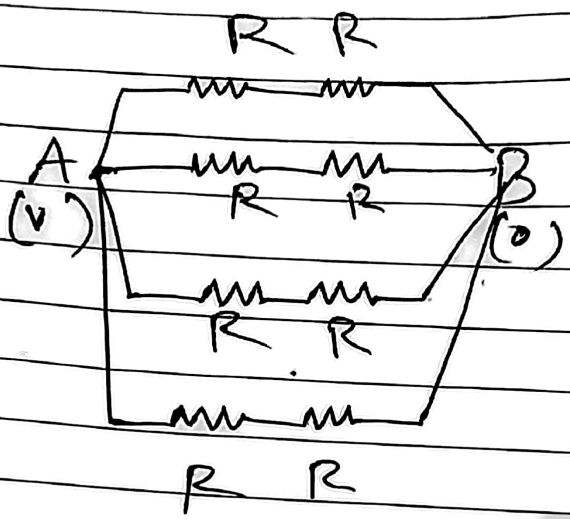
or $Re = \frac{R}{n}$ $\quad Re = \frac{R}{3}$

② 

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Solution





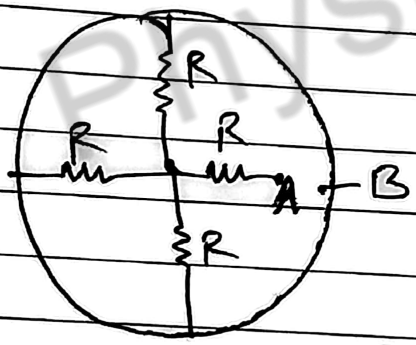
$$\frac{1}{Re} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{2R} + \frac{1}{2R}$$

$$\frac{1}{Re} = \frac{4}{2R}$$

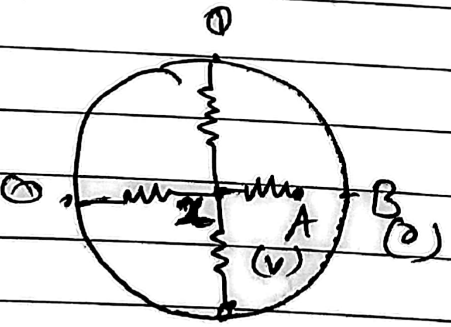
$$Re = \frac{R}{2}$$

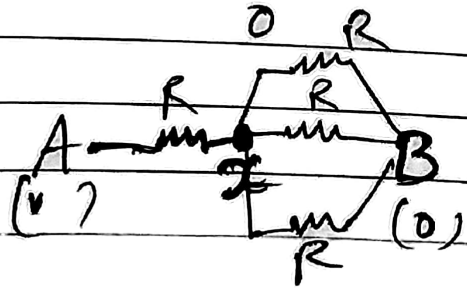
(Or $Re = \frac{R}{n} = \frac{2R}{4} = \frac{R}{2}$)

(3)



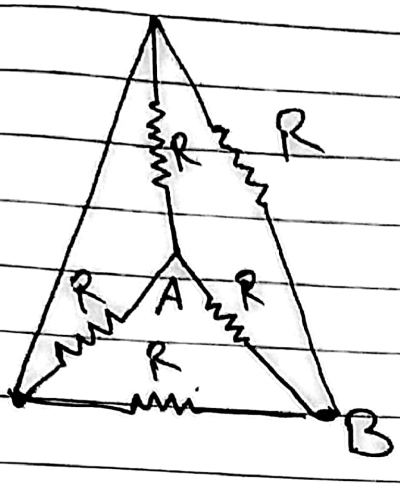
Solution



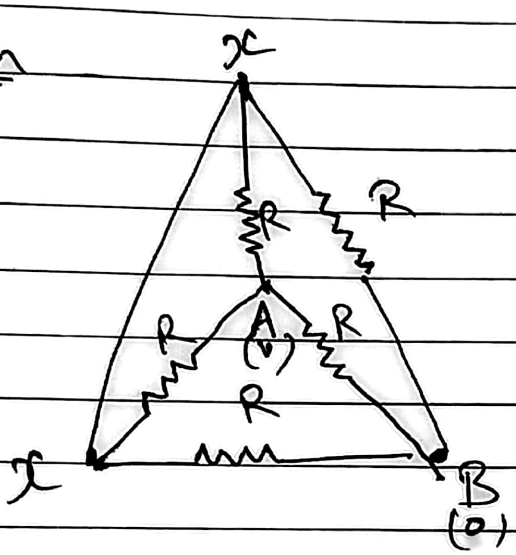


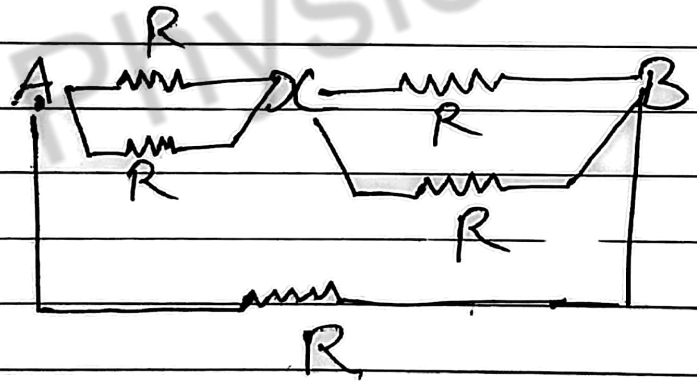
$$Re = R + \frac{R}{3} = \frac{4R}{3}$$

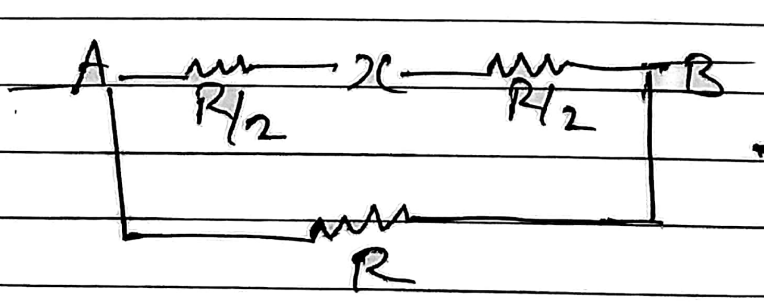
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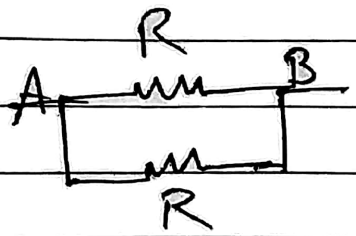
Solution







⇒

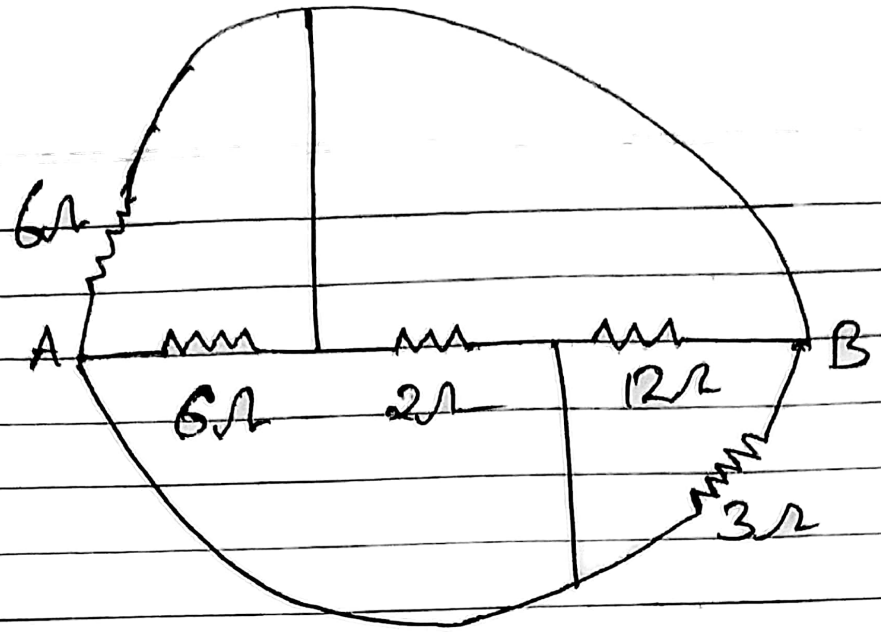


⇓

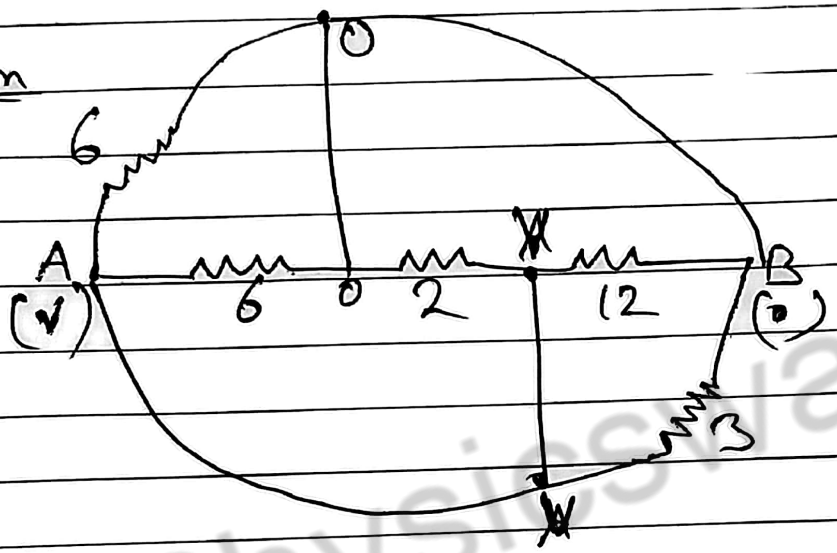


Physicswallah

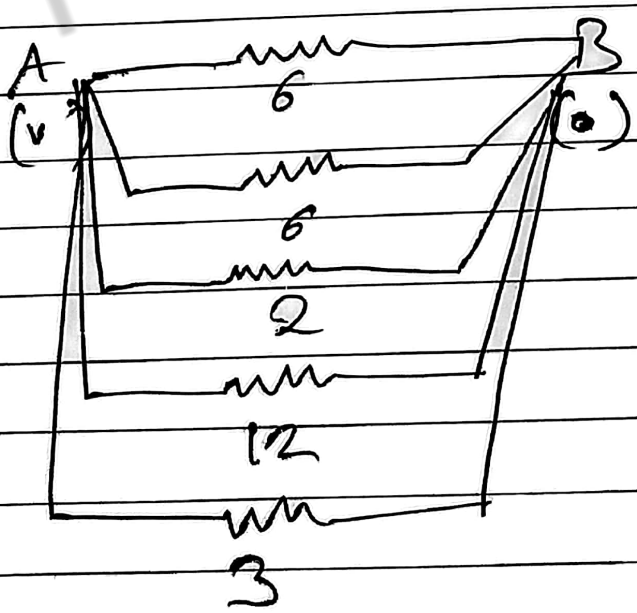
(5)



Solution



Physicswallah



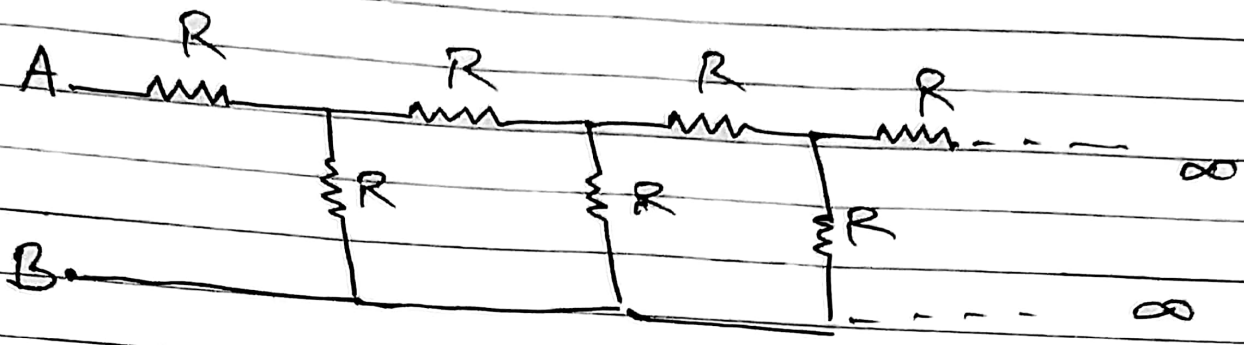
$$\frac{1}{Re} = \frac{1}{6} + \frac{1}{6} + \frac{1}{2} + \frac{1}{12} + \frac{1}{3}$$

$$\frac{1}{Re} = \frac{2+2+6+1+4}{12}$$

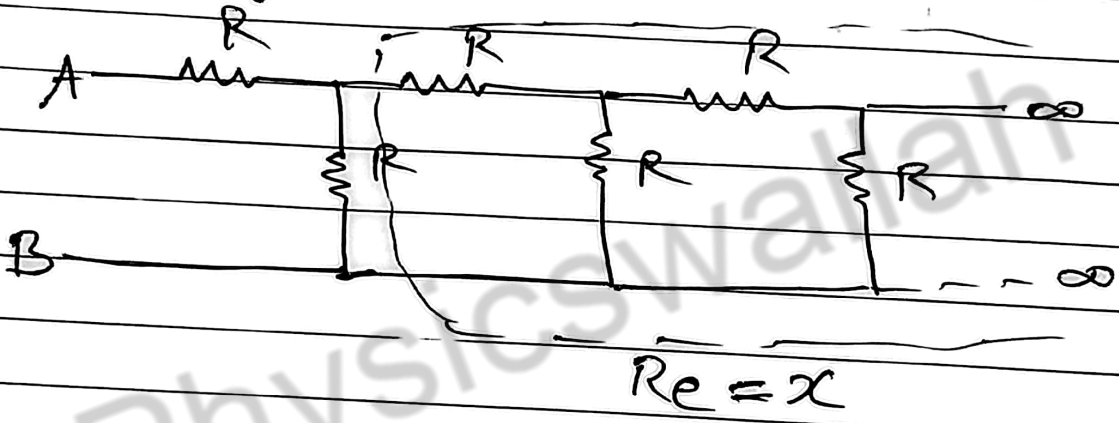
$$\frac{1}{Re} = \frac{15}{12} \frac{5}{4}$$

$$Re = \frac{4}{5}$$

II) Infinite Network:



Solution Let equivalent Resistance About AB be 'x' of whole circuit ⟶ x

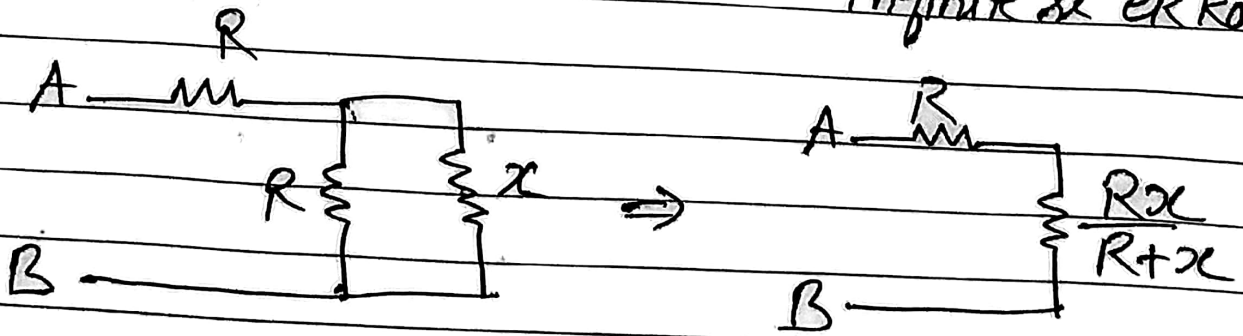


ek unit chor ke (Jo repeat ho rahi ho) remaining ka equivalent ⟶ x

⇓

infinite ka x

infinite se ek kam ka x



now eq AB ⇒ $R_e = R + \frac{Rx}{R+x}$

Physicswallah

and eq AB is x

$$R_e = R + \frac{Rx}{R+x}$$

$$x = R + \frac{Rx}{R+x}$$

$$x = \frac{R(R+x) + Rx}{(R+x)}$$

$$Rx + x^2 = R^2 + Rx + Rx$$

$$x^2 - Rx - R^2 = 0$$

$$x = \frac{R \pm \sqrt{R^2 + 4R^2}}{2}$$

$$x = \frac{R \pm R\sqrt{5}}{2}$$

$$x = \frac{R - R\sqrt{5}}{2} \qquad x = \frac{R + R\sqrt{5}}{2}$$

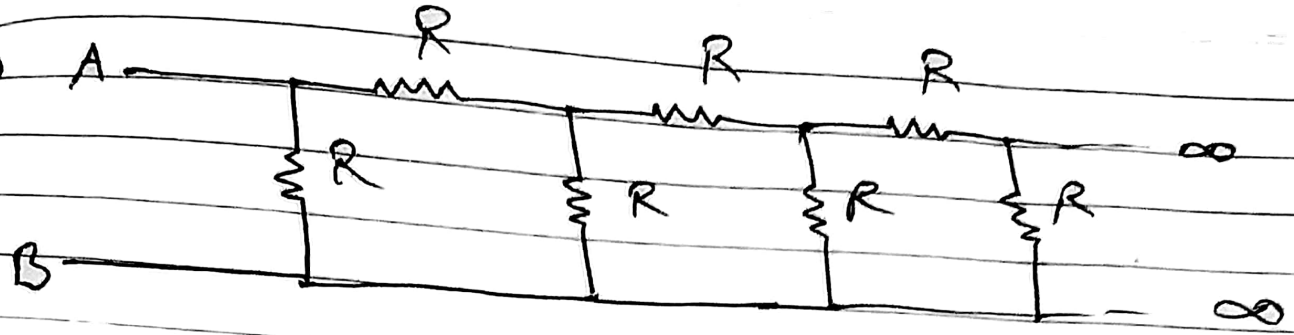
$= -ve$
not possible

$$x = R\left(\frac{1+\sqrt{5}}{2}\right)$$

$$\boxed{R_e = R\left(\frac{1+\sqrt{5}}{2}\right)}$$

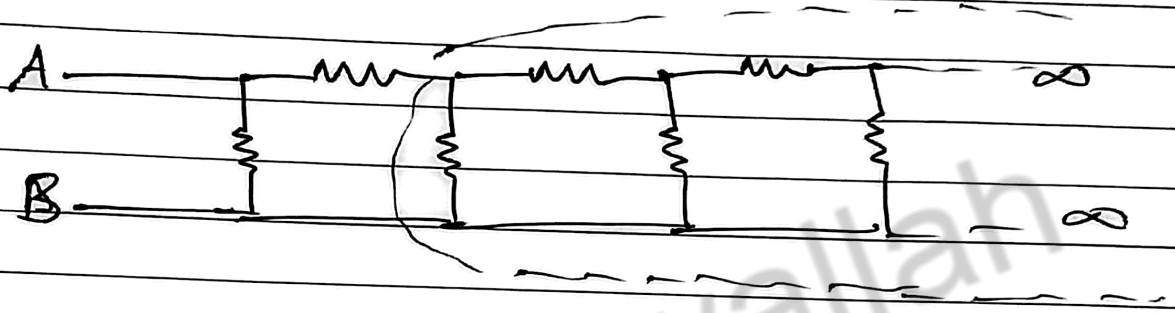
PhysicsWallah

(2)



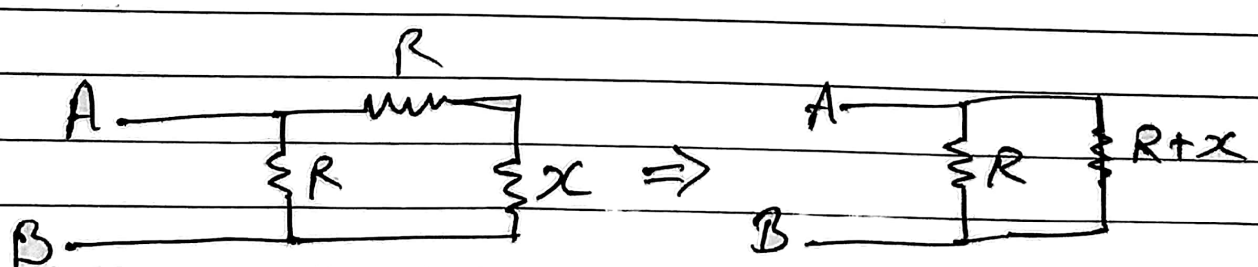
Solution:

Let equivalent Resistance between AB be x
for whole circuit → x



ek unit chor k (Jo repeat ho rahi ho)
remaining ka eq → x

∞ ka x
to ∞ me se ek hata kar bhi x



$$A\text{-}B\ R_e = \frac{R(R+x)}{R+(R+x)}$$

& A-B Req is x as assumed

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$$R_e = \frac{R(R+x)}{R+(R+x)}$$

$$\downarrow$$

$$x = \frac{R(R+x)}{R+(R+x)}$$

$$x(2R+x) = R^2 + Rx$$

$$2Rx + x^2 = R^2 + Rx$$

$$x^2 + Rx - R^2 = 0$$

$$x = \frac{-R \pm \sqrt{R^2 + 4R^2}}{2}$$

$$x = \frac{-R \pm R\sqrt{5}}{2}$$

$$x = \frac{-R - R\sqrt{5}}{2} \qquad x = \frac{-R + R\sqrt{5}}{2}$$

$$= -ve$$

not possible $\qquad x = \frac{R(\sqrt{5}-1)}{2}$

$$\boxed{R_e = \frac{R(\sqrt{5}-1)}{2}}$$

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